

HW-3

8.2. $H(z) = 8 \left[\frac{1-4z^{-1}}{1-\frac{1}{6}z^{-1}} \right]$

(a) $\hat{H}(z) = \ln(H(z)) = \ln 8 + \ln(1-4z^{-1}) * -\ln(1-\frac{1}{6}z^{-1})$

而 $\ln(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n} \quad |x| < 1$

对于 $\ln(1-4z^{-1}) = -\sum_{n=1}^{\infty} \frac{4^n}{n} z^{-n}$; 对于 $\ln(1-\frac{1}{6}z^{-1}) = -\sum_{n=1}^{\infty} \frac{(\frac{1}{6})^n}{n} z^{-n}$

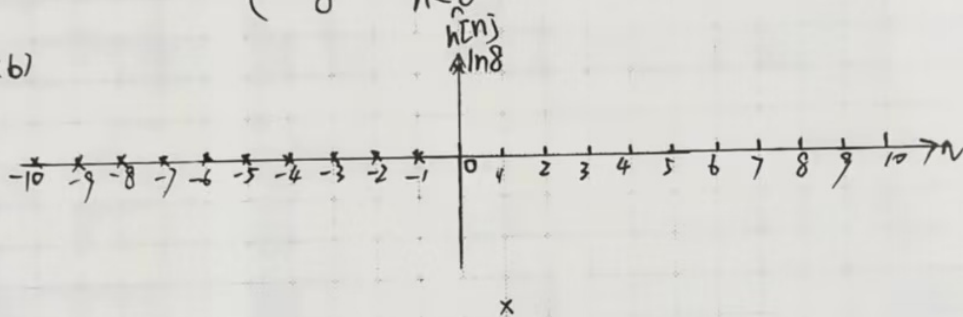
$|4z^{-1}| < 1$ 且 $|\frac{1}{6}z^{-1}| < 1$ 即 $|z| > 4$ 且 $|z| > \frac{1}{6}$

$\hat{H}(z) = \ln 8 - \sum_{n=1}^{\infty} \frac{4^n}{n} z^{-n} + \sum_{n=1}^{\infty} \frac{(\frac{1}{6})^n}{n} z^{-n}$

与 $H(z) = \sum_{n=-\infty}^{+\infty} \hat{h}[n] z^{-n}$ 可得:

$\hat{h}[n] = \begin{cases} \ln 8 & n=0 \\ \frac{(\frac{1}{6})^n - 4^n}{n} & n>0 \\ 0 & n<0 \end{cases}$

(b)



(c) $c[n] = \frac{\hat{h}[n] + \hat{h}[-n]}{2} = \begin{cases} \ln 8 & n=0 \\ \frac{(\frac{1}{6})^{|n|} - 4^{|n|}}{2|n|} & n \neq 0 \end{cases}$ (后面是巨大的负值, 不便展示)

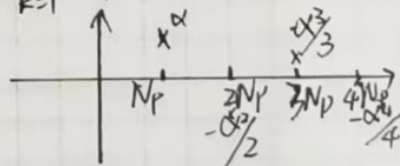
8.8. $x[n] = \delta[n] + \alpha \delta[n-N_p] \quad |\alpha| < 1$

(a) $X(z) = 1 + \alpha z^{-N_p} \quad \hat{X}(z) = \ln X(z) = \ln(1 + \alpha z^{-N_p})$

利用 $\ln(1+z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} z^k \quad |z| < 1$

$\hat{X}(z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \alpha^k z^{-kN_p} \quad \hat{X}[n] = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \alpha^k \delta[n-kN_p]$

$\hat{X}[n] = \begin{cases} \frac{(-1)^{k+1} \alpha^k}{k} & n = kN_p \quad k=1, 2, 3, \dots \\ 0 & \text{其他} \end{cases}$



$$(b) \quad c[n] = \frac{x[n] + x[n]}{2} = \begin{cases} \frac{(1-1)^{k+1} \alpha^k}{2k} & n = \pm kN_p \quad k=1, 2, 3, \dots \\ 0 & \text{其他} \end{cases}$$

$$(c) \quad N_p = N/6 \quad 8.58(c): \quad \hat{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \hat{x}[k] e^{j \frac{2\pi}{N} kn} \quad 0 \leq n \leq N-1$$

$$\hat{x}[k] = \ln X[k] = \ln (1 + \alpha e^{-j \frac{2\pi}{N} k N_p})$$

$$\text{当 } N_p = N/6 \quad \therefore e^{-j \frac{2\pi}{N} k N_p} = e^{-j \frac{2\pi}{6} k} = e^{-j \frac{\pi}{3} k}, \text{ 这是周期为6的函数}$$

$$\hat{x}[n] = \sum_{r=-\infty}^{+\infty} \hat{x}[n+rN] \quad \therefore \hat{x}[n] \text{ 只在 } n = N_p, 2N_p, \dots \text{ 处有值, 而 } N = 6N_p$$

在 $0 \leq n \leq N-1$ 范围内

$$\hat{x}[N_p] = \hat{x}[N_p] = \alpha \quad \hat{x}[2N_p] = \hat{x}[2N_p] = -\frac{\alpha^2}{2}$$

$$\hat{x}[3N_p] = \hat{x}[3N_p] = \frac{\alpha^3}{3} \quad \hat{x}[4N_p] = \hat{x}[4N_p] = -\frac{\alpha^4}{4}$$

$$\hat{x}[5N_p] = \hat{x}[5N_p] = \frac{\alpha^5}{5} \quad \hat{x}[6N_p] = \hat{x}[6N_p] = -\frac{\alpha^6}{6}$$

$$\text{对于 } n \in [0, N-1] \quad \hat{x}[n] = \hat{x}[n] + \sum_{r=1}^{\infty} \hat{x}[n+rN] = \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \alpha^m \delta(n - mN_p \bmod N)$$

若不能整除, 则会导致频谱泄漏和混叠失真

$$(d) \quad \tilde{c}[n] = \frac{\log |X[k]|}{N} \sum_{k=0}^{N-1} \log |X[k]| e^{j \frac{2\pi}{N} kn} \quad 0 \leq n \leq N-1$$

$$X[k] = 1 + \alpha e^{-j \frac{2\pi}{N} k N_p}$$

$$\log |X[k]| = \frac{1}{2} \log (1 + \alpha^2 + 2\alpha \cos(\frac{2\pi}{N} k N_p))$$

$$= - \sum_{m=1}^{\infty} \frac{(-\alpha)^m}{m} \cos(\frac{2\pi}{N} k m N_p)$$

$$\tilde{c}[n] = - \sum_{m=1}^{\infty} \frac{(-\alpha)^m}{m} \cdot \frac{1}{N} \sum_{k=0}^{N-1} \cos(\frac{2\pi}{N} k m N_p) e^{j \frac{2\pi}{N} kn}$$

$$\therefore \tilde{c}[n] = \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{2m} \alpha^m [\delta(n - mN_p \bmod N) + \delta(n + mN_p \bmod N)]$$

$$(e) \quad N - N_p \geq N_p \quad \therefore N \geq 2N_p$$